RESUME - Dr Jamie Gabriel

5 Prince St, Fennell Bay NSW 2283 | 0490 896 610 | jgab3103@gmail.com | jamiegabriel.org

---

SUMMARY

- Senior technology and data executive with extensive experience leading enterprise data strategy, information governance, software engineering and digital transformation across complex public sector, corporate and university environments.
- Over three decades of hands-on experience in software development, database design, systems architecture and emerging technologies, including artificial intelligence, machine learning, cloud computing and advanced analytics.
- Proven track record designing and delivering enterprise technology solutions, data governance frameworks and complex information systems, encompassing the full software development lifecycle, data integration, information security, privacy, regulatory compliance and organisational change.
- Experienced leader of multidisciplinary technical and professional teams, with a strong record of establishing collaborative partnerships, mentoring technical specialists and translating complex business requirements into practical, scalable technology solutions.
- Extensive experience in academic research, education and applied technology development, combining advanced technical expertise with an interdisciplinary approach to problem-solving, innovation and knowledge sharing.
- Strategic and collaborative leader with demonstrated ability to advise senior executives, manage complex programs and deliver sustainable organisational improvements, from initial strategy and solution design through to implementation and business-as-usual operations.

RECENT PROFESSIONAL EXPERIENCE

Teaching / Freelance Software Developer (Learning Design) **Aug 2025 – Present**

Achievements:

- Design and develop interactive educational software, digital learning tools and web-based applications supporting music education, including ear training, music notation, harmony, rhythm, improvisation and composition.
- Apply software engineering, learning design and educational technology principles to create engaging, accessible applications that support independent learning and accommodate diverse abilities and learning needs.
- Design, develop and publish interactive learning applications and digital teaching resources through my website, jamiegabriel.org, integrating interactive interfaces, audio synthesis, music theory and real-time feedback.
- Undertake independent software development projects, including the design and development of BeatMatrix, an iOS application for advanced rhythmic training and complex musical metres.
- Manage the full software development lifecycle, including requirements analysis, solution architecture, application design, development, testing, deployment, version control and ongoing maintenance, applying modern software engineering practices and development frameworks.
- Design, develop and manage advanced database systems, including relational and NoSQL architectures, data modelling, query optimisation, data integration and performance management, supporting scalable applications and complex information requirements.

- Provide consulting and technical advice on artificial intelligence, machine learning and emerging technologies, including their practical implementation and adoption.

Chief Data Officer, Cancer Institute NSW **Dec 2021 - August 2025**

Achievements:

- Led the development and implementation of the Institute's enterprise Data Strategy and governance framework, aligning information management, data architecture, governance, risk and technology initiatives with organisational priorities and broader NSW Health programs.
- Established and embedded enterprise data governance arrangements, bringing together data, analytics, IT and business stakeholders to clarify ownership, accountability, standards and processes for the management and use of organisational information.
- Held enterprise responsibility for the governance and stewardship of the Institute's data assets, including sensitive health datasets, with a strong focus on appropriate access, security, privacy, information sharing and risk management.
- Built strong relationships with senior stakeholders across NSW Health and the broader research sector, and led the implementation of a purpose-built research collaboration platform, establishing governance arrangements to support secure and compliant information sharing.
- Led a significant enterprise data architecture and systems transformation, establishing scalable cloud-based infrastructure and improving the organisation's ability to manage, integrate, access and utilise information while maintaining appropriate security and governance controls.
- Provided strategic and technical advice to executives and senior stakeholders on information governance, data management, privacy, risk, architecture and legislative requirements, translating complex technical issues into practical organisational solutions.
- Led multidisciplinary teams and cross-organisational initiatives, establishing clear governance structures, operating practices and stakeholder responsibilities to support the transition from strategic design into sustainable business-as-usual operations.

Manager, Data & Research Governance, & NSW Cancer Data Custodian Cancer Institute NSW **Oct 2017 - Dec 2021**

Achievements:

- Managed the data and research governance function of the Institute, and introduced multiple solutions related to the tracking of data assets and access, data element lineage, master and reference data, research programs of work, and privacy and security incident / risk management.
- Oversaw the creation of new processes and systems for all aspects of Institute data analytics across Divisions, including the development of standard coding practices to underpin data engineering, analytics, data asset onboarding, code management, and introduced a suite of new practices to support code review, SME collaboration, version control management, and the development of common reference data sets.
- Upgraded all data governance and system architecture compliance documentation and processes, and implemented metrics and reporting to track data custodianship approvals, and integrated all governance processes with version control and implemented data services for monitoring and metrics.
- Developed the Institute Data Strategy and managed its implementation across the domains of enterprise architecture, governance, analytics and reporting.

Lead, Data Projects, Department of Education **2015 - 2017**

Achievements:

- Developed the organisation's data, analytics and reporting capability from the ground up, establishing governance processes, infrastructure and operating practices to support the consistent management, analysis and reporting of enterprise information.
- Established an analytics and data-sharing partnership model with NSW Government and Australian Government agencies, developing collaborative arrangements for the secure and appropriate sharing, integration and use of information across organisational boundaries.
- Led an interdisciplinary team through several major software upgrades, system migrations and legacy system remediation projects, ensuring business continuity while improving system capability, information quality, security and operational efficiency.
- Led the design and implementation of multiple enterprise software solutions, including funding calculation, reporting and automated briefing systems, replacing manual processes with controlled, auditable and repeatable workflows.
- Oversaw the design and deployment of systems supporting the calculation and distribution of more than $200 million in NSW early childhood funding, establishing robust business rules, data controls and processes to ensure accuracy, accountability and appropriate use of public funds.
- Worked closely with senior executives, policy teams, technology teams and external government agencies to translate complex business and information requirements into practical governance, technology and operational solutions.
- Established processes for the ongoing management, validation and quality assurance of critical datasets, supporting reliable reporting, funding decisions and government accountability.
- Provided strategic and technical advice to senior stakeholders on data management, information sharing, systems architecture, analytics and digital transformation, balancing operational requirements with governance, security and risk considerations.

Macquarie University / Teaching & Projects **2009 - 2015**

Achievements:

- Delivered numerous learning and teaching analytics software and scientific research projects (i.e. AI and machine learning driven predictive models to allow real-time learning interventions and curriculum design);
- Oversaw the design and implementation of the SALT project, a high-volume workflow engine for tracking and analysing student waivers and special consideration applications, integrated with the main student system and supported by dedicated data APIs for advanced analytics and transaction-level monitoring.

General Manager, Faculty of Pharmacy, University of Sydney **2006 - 2009**

Achievements:

- Led a major strategic faculty transformation program, overseeing organisational change, infrastructure development, procurement and implementation across teaching and research environments within a complex university setting.
- Managed the planning, procurement and implementation of several new teaching and research laboratories, coordinating academic, technical, procurement, facilities and external stakeholders and ensuring delivery against operational, safety, budget and compliance requirements.
- Designed and implemented a custom enterprise asset management system to establish centralised visibility and governance of medical and scientific equipment, including asset registers, ownership, lifecycle management, maintenance, support scheduling and replacement planning.
- Established asset lifecycle processes and controls that improved accountability for high-value equipment and provided management with greater visibility of asset status, utilisation, maintenance requirements and future investment needs.
- Led a major uplift of the Faculty's technology and research computing environment, including network capacity and high-performance CPU infrastructure, to support computational chemistry research and emerging digital research requirements.
- Provided strategic advice to academic and professional leadership on technology, infrastructure, procurement, asset management, risk and operational planning, translating complex technical requirements into sustainable business solutions.
- Worked across academic, professional and technical teams to develop enterprise processes, systems and governance arrangements that supported the Faculty's long-term teaching, research and operational objectives.

Education:
PhD University of Technology
Topic Area:
Data Science / Computer Science: (Designing search and retrieval models for domain-specific MusicXML)

Other Relevant Skills:

Records, Information and Data Governance

Extensive experience developing and implementing enterprise data and information governance frameworks, including data ownership and custodianship, information asset registers, access governance, metadata, data lineage, classification, governance approvals, privacy and security controls, risk management, and compliance reporting. Experience establishing governance processes across complex, highly regulated organisations and integrating governance requirements with technology, architecture and business processes.

Governance, Risk and Compliance

Extensive experience designing governance, assurance and risk-management processes for sensitive and high-value information. Experienced in developing policies, standards, approval workflows, governance documentation, reporting and assurance mechanisms, and providing strategic advice to senior executives on information management, privacy, security, legislative and regulatory requirements.

Enterprise Solution Architecture

Extensive experience delivering large-scale application and ICT infrastructure projects with a data and information focus, including enterprise data architecture, data pipelines, cloud deployments, streaming and elastic data storage, system integration, legacy system transformation and migration, and third-party contract management.

Software Engineering and Data Science

Extensive experience building and deploying data science, machine learning and software solutions, with strong technical design skills developed through managing projects involving systems architecture, database design and implementation.

Information Systems, Metadata and Data Lifecycle Management

Extensive experience designing systems and processes to support the end-to-end management of information assets, including asset registration, ownership, access, metadata, lineage, version control, data quality, retention considerations, security and lifecycle management. Experienced in establishing consistent information standards and processes across distributed business environments.

Technical Skills

Programming languages: Over three decades of software development experience using R, Python, C, Objective-C (iOS), C++, Java, JavaScript/JSX/TypeScript, Angular, React and Haskell, with extensive experience designing and building desktop, mobile and networked applications.

Software Frameworks and Database Administration: Extensive experience with Django, React, Docker, D3.js, Angular.js, Ruby on Rails, Shiny, Apache Kafka, Apache Airflow, MongoDB, Elasticsearch, Neo4j, PostgreSQL, MySQL, Node.js, Electron, Apple iOS, RStudio, Jupyter, TensorFlow, Dash and Mathematica.

OS and Networks: Advanced Linux, including shell scripting, task management, server configuration and administration; advanced Windows and macOS.

Referees:

Referees Available on Request.

jamie.gabriel@health.nsw.gov.au